CIS 106 – Loops Part 2

For each problem prepare an IPO chart. Then write the code for each. Save the IPO within this document and upload to your repository. After code is complete upload the files (.py) to your repository. Paste the link to your repository into the assignment completion link in Blackboard.

1. Allow the user to enter a principle amount and interest rate repeatedly (need a loop to control the program execution). Compute the annual interest (principle x rate). Compute ending balance to be principle (beginning balance + interest). Display year, beginning balance and ending balance for each of the 5 years. Display the accumulated interest for the 5 years. Note: the new balance by year (this will be the principle for the following year. Format the output.

INPUT	PROCESS	OUTPUT
Principal amount and interest rate (loop)	Compute the annual interest (principal x rate). Compute the ending balance to be principle (beginning balance + interest).	Display year, beginning balance, and ending balance for each of the 5 years.
		Display the accumulated interest for the 5 years.

Example:

Enter principle amount: 10000.00

Enter interest rate: 0.10

Year	Beginning Balance	Ending Balance
1	$10,000.00	$11,000.00
2	$11,000.00	$12,100.00
3	$12,100.00	$13,310.00
4	$13,310.00	$14,641.00
5	$14,641.00	$16,105.00

Total interest earned: $6,156.00

2. The Fibonacci sequence is a sequence of natural order. The sequence is:
   1, 1, 2, 3, 5, 8 etc
   Use of for loop compute and display the first 20 numbers in the sequence. Hint: start with 1 , 1.

INPUT	PROCESS	OUTPUT
Fibonacci sequence	f(n-1)+f(n-2) Enter 1 as first number	Display the first 20 numbers in the sequence
[redacted]		

3. Create a text file that contains employee last name and salary. Read in this data. Determine the bonus rate based on the chart below. Use that rate to compute bonus. For each line display the employee last name, salary and bonus. After the loop display the sum of all bonuses paid out.

INPUT	PROCESS	OUTPUT
f = open("PS7P3.txt", "r") #initialize counters and sums to 0 - typical statements before the loop when summing and/or counting is required c = 0.0 totp_ex = 0.0 item = str(f.readline().rstrip('\n')) while item !="":	Determine the bonus rate based on the chart below. Yearly_salary greater than or equal to 100,000, bonus_rate=.20 Yearly_salary less than 100,000 but greater than 50,000, bonus_rate=.15 Yearly_salary less than 50,000, bonus rate = .10	For each line, display the employee's last name, salary, and bonus. Display the sum of all bonuses paid out.

[redacted]	[redacted]	

Salary	Bonus Rate
100,000.00 and up	20%
50,000.00	15%
All other salaries	10%

4. Create a text file with item, quantity and price. Read through the file one line at a time. Compute the extended price (quantity x price). For each line display the item, quantity, price and extended price. After the loop display the sum of all the extended prices, the count of the number of orders and the average order.

INPUT	PROCESS	OUTPUT

f = open("PS7P4.txt", "r")
`#initialize counters and sums to 0 - typical statements before the loop when summing and/or counting is required`
`c = 0.0`
`totp_ex = 0.0`
`item = str(f.readline().rstrip('\n'))`
` while item !="":`
	Compute the extended price (quantity x price)	For each line, display the item, quantity, price, and extended price.
		Display the sum of all the extended prices, the count of the number of orders, and the average order.

5. Create a text file with student last name, district code (I or O) and number of credits taken. Compute tuition owed (credits taken x cost per credit). Cost per credit for in district students (district code I) is 250.00. Out of district students pay 500.00 per credit. For each line display student last name, credits taken and tuition owed. After the loop display sum of all tuition owed and the number of students.

INPUT	PROCESS	OUTPUT

f = open("PS7P5.txt", "r")
`#initialize counters and sums to 0 - typical statements before the loop when summing and/or counting is required`
`c = 0.0`
`totp_ex = 0.0`
`item = str(f.readline().rstrip('\n'))`
`while item !="":`
	Compute tuition owed (credits taken x cost per credit). Cost per credit for in district students (district code I) is 250.00. Out of district students pay 500.00 per credit.	Display student last name, credits taken and tuition owed. After the loop display sum of all tuition owed and the number of students.

Example file

Jones

I

12

Adams

I

10

Baker

O

12

Smith

O

16